Capacity by construction: CVRP clustering as maximum independent set on a neutral-atom register

DuQuantum 2026 — Pasqal challenge, Track A

Christopher Thai, Alex Wang, Kevin Nguyen

October 25, 2026

1. Problem and decomposition

Customers $i \in \{0, \dots, N-1\}$ with positions P_i and demands d_i; K identical vehicles of capacity Q at a fixed depot (50, 50); Euclidean costs. Decision variables of the clustering step: $x_{ik} = 1$ iff customer i is served by vehicle k. Constrained clustering model:

$$\min \sum_k \sum_{i<j} D_{ij}\, x_{ik} x_{jk} \quad \text{s.t.} \quad \underbrace{\sum_k x_{ik} = 1 \;\forall i,}_{\text{equality}} \qquad \underbrace{\sum_i d_i x_{ik} \le Q \;\forall k\,.}_{\text{inequality}}$$

Routes are then obtained per cluster by an exact TSP (cluster-first / route-second). Instances follow Table 1 of the brief; each is drawn from a seed and shipped with a feasibility certificate obtained by an exact bin-packing search (draws without a packing are rejected, so generation stays a deterministic function of the seed). Load ratios ρ are 0.55–0.93 and $d_{\max}/Q$ 0.30–0.57 (`instances/references.csv`).

References are proven optima on every tier. A Held–Karp dynamic programme over all 2^N customer subsets gives the exact depot-anchored tour of every possible cluster at once; a set-partitioning integer programme over the *complete* pool of capacity-feasible subsets (scipy/HiGHS, 15–32 768 columns) then gives the CVRP optimum and, with the pairwise-distance block cost, the optimum of the clustering model. On S and M a brute-force enumeration of all partitions agrees and counts distinct optima (vehicle relabellings identified). Tier L takes ≈2 s. Decomposition gaps (model optimum routed exactly vs CVRP optimum) are 0–16 % on S, 3–13 % on M, 1–12 % on L.

2. What the register minimises, and what that rules out

Once the amplitude is back at zero, Eq. (1) of the brief is the classical cost $E(n) = -\delta \sum_a n_a + \sum_{a<b} \frac{C_6}{r_{ab}^6} n_a n_b$. Through the laser we control one global δ: every variable has the *same* linear coefficient. Through the positions we control the couplings, but they are all positive and fall as $1/r^6$; nothing is ever exactly uncoupled, and with 5 µm minimum separation and 38 µm radius the useful band is roughly 5–10 µm (doubling the distance divides the coupling by 64). At $\Omega_{\max} = 12.57$ rad/µs the blockade radius is $(C_6/\Omega)^{1/6} = 6.4$ µm. This cost is maximum independent set (MIS) on a unit-disk graph.

Two consequences shaped everything. (i) *Linear terms carry no information* without the DMM. On the one-hot manifold $\sum_k x_{ik} = 1$ that is harmless: any per-customer coefficient gives $\sum_k c_i x_{ik} = c_i$, a constant, so expanding a quadratic load penalty $\sum_k (L_k - T)^2$, $L_k = \sum_i d_i x_{ik}$, leaves only $2\sum_k \sum_{i<j} d_i d_j x_{ik} x_{jk}$: heavy customers repel within a vehicle, no slack, no linear term — but a *dense* coupling pattern. (ii) *Dense couplings cannot be realised*: an atom can have about three strongly coupled neighbours whose mutual couplings are weak, and mutually conflicting atoms must fit inside one blockade disc. Any model needing more is an approximation whose error must be measured; we measured it by brute-forcing the ground state of the *realised* register cost (`embedding.classical_ground_states`, `manifold_check`) before any emulation.

Variable accounting (Derive 4.1)

encoding	count	S	M	L	fits 80?
assignment variables only	NK	8	30	60	yes

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+ binary slack, $\lceil \log_2(Q{+}1) \rceil$ bits/vehicle	$NK + K\lceil \log_2(Q{+}1) \rceil$	14	45	80	S, M; L on paper only (slack couples to every assignment atom)
one-hot by blockade, no slack	NK	8	30	60	yes, but unembeddable at M/L (§5)
ball packing (this work)	#candidates $\leq N$	2–4	5–6	6–10	yes, exact state-vector emulation on every tier

3. The model: capacity by construction, packing by blockade

Instead of penalising the inequality, we changed what the register is asked for. For every customer centre, take the nearest customers in order of distance until the load would exceed $f\,Q$ ($f = 0.85$); that set is a **candidate cluster, capacity-feasible by construction**. Near-duplicate candidates (Jaccard overlap > 0.6) are thinned, because candidates that pairwise overlap must sit inside one blockade disc and cliques larger than three or four cannot. One atom per candidate; two atoms are placed inside the blockade radius iff their candidates share a customer. With a uniform positive final detuning the ground state is a **maximum set of pairwise-disjoint compact clusters**. Two candidates overlap only when their centres are close on the map, so the conflict graph is close to a unit-disk graph of the map itself: the register is the map at scale, which is what a neutral-atom array embeds best. There is no penalty weight to tune; the knobs are f and the Jaccard threshold.

Classical completion (stated, identical for quantum, oracle and random shots). The device's chosen candidates are kept intact; uncovered customers are then placed by one of two completions, both reported: *greedy* (each leftover to the chosen cluster with the cheapest marginal pairwise distance that has slack) and *exact* (every assignment of the leftovers to the K vehicles is enumerated with the exact tour cost of each resulting cluster from the Held–Karp table, optimal given the chosen candidates whenever ≤ 8 customers are uncovered). The covered fraction is reported for every shot.

Selection rule (stated). The answer is the device's best *raw-feasible* shot — an independent set of at most K candidates — by exact routed cost after completion (if the device lit more than K candidates, every K-subset is completed and the cheapest kept). Shots with overlapping candidates are only a fallback, resolved greedily and flagged `repaired`. We also report the best over *all* shots after completion (`cvrp_gap_pct_any`), because that is what a purely classical completion would claim.

4. Embedding and pulse

Embedding (A3). Positions are fitted by least squares in log-distance (L-BFGS with analytic gradients, restarts from the candidate centres): conflict pairs at 6.0 µm (18.6 rad/µs), non-conflict pairs beyond 9.5 µm (1.2 rad/µs), then a hard projection onto the envelope (≥ 5 µm, ≤ 38 µm radius), which pulser's `AnalogDevice` validates. Fidelity formulas: rel_err $= \sqrt{\sum_{\text{wanted}}(U-T)^2 / \sum T^2}$, log-RMS of $\ln(U/T)$ over wanted pairs, the leakage ratio $\sum_{\text{unwanted}} U / \sum_{\text{wanted}} T$, and — the one that predicts the physics — the graph-level count of **violable edges** (conflict pairs realised below δ_f) and **spurious edges** (non-conflicts above it). On the 15 instances the registers have 0–8 violable and 0 spurious edges out of 0–26 overlap edges (`results/results.csv`, columns `violable_edges_levels`).

Final detuning and amplitude. $\delta_f = \sqrt{18.6 \times 1.2} = 4.7$ rad/µs sits at the geometric mean of the conflict coupling and the floor, so a conflict pair costs more than two excitations gain and a non-conflict pair costs less. $\Omega_{\max} = 4$ rad/µs gives a blockade radius of 7.7 µm, between the two design distances. The detuning starts at $-1.5\,\delta_f$ so the initial state is the empty register.

Pulse (A4). Baseline: three-point interpolated amplitude $0 \to \Omega_{\max} \to 0$, linear detuning, 4000 ns (device maximum 6000 ns). we scored pulses by the metrics that reward good shots — feasibility rate, $P(\text{shot is a maximum packing})$ and the best gap — not by mean energy. The sweep over duration {1, 2, 4, 6} µs, $\Omega_{\max} \in \{2, 4, 8\}$ rad/µs and a `slow-end` detuning that spends 70 % of the sweep on the positive side (240 emulations, `results/pulse_sweep.csv`, Fig. 2) gives a clean adiabaticity curve: $P(\text{max packing per shot})$ rises monotonically with duration, on L from 0.06 at 1 µs to 0.45 at 4 µs and 0.68 at 6 µs, on M from 0.27 to 0.74, while the **best gap over 500 shots is the same for every pulse** — even the 1 µs sweep reaches a maximum packing at least once. $\Omega_{\max} = 8$ is worst at every duration (its blockade radius, 6.9 µm, sits too close to the 6 µm conflict distance and blurs the conflict/non-conflict distinction); 2 and 4 rad/µs are equivalent. `slow-end` helps on L at 6

µs (0.68 vs 0.56 at $\Omega = 2$) and is neutral on M. Raw feasibility barely depends on the pulse (0.57–0.83): diabatic sweeps still return independent sets, just smaller ones. Final choice: 6 µs, $\Omega_{\max} = 2$, `slow-end`.

5. Results

Emulation: pulser 1.9.1, `AnalogDevice` (automatic trap layout), exact QuTiP state-vector backend (≤ 12 atoms, so no bond dimension is involved), 500 shots per run, 3 runs per instance, 20 instances per tier. Every (instance, sampler, run) is sampled **once** and the same shots are post-processed with both completions, so the two halves of the table differ only in the classical step. Means over instances; "any-shot" is the best over all shots after completion. [QuTiP rows: FILL from `results/final.csv` after `run_final.py --samplers qutip`; classical-sampler rows below are final.]

tier	atoms	sampler	raw feas.	P(max packing/shot)	gap, greedy completion	gap, exact completion
S	2–4	QuTiP sweep	[F]	[F]	[F]	[F]
S	2–4	ideal MIS of the target graph	0.73	0.73	6.8 %	3.9 %
S	2–4	random bitstrings	0.69	0.26	5.4 %	1.5 %
M	4–8	QuTiP sweep	[F]	[F]	[F]	[F]
M	4–8	ideal MIS	0.92	0.92	5.1 %	2.2 %
M	4–8	oracle (exact ground state of realised register)	0.60	0.60	9.9 %	4.1 %
M	4–8	random bitstrings	0.22	0.06	5.4 %	0.0 %
L	5–12	QuTiP sweep	[F]	[F]	[F]	[F]
L	5–12	ideal MIS	1.00	1.00	9.7 %	3.0 %
L	5–12	oracle	0.80	0.80	12.6 %	7.8 %
L	5–12	random bitstrings	0.15	0.03	9.9 %	1.7 %

Four observations. **The classical completion, not the sampler, sets the final gap.** The exact completion lowers every row by 3–8 points, and under it random bitstrings reach 0–1.7 %: with ≤ 8 uncovered customers the exhaustive completion is itself a near-complete solver, and a random shot (coverage 0.4–0.6) hands it most of the problem. Under the greedy completion the ideal register and random bitstrings tie (9.7 % vs 9.9 % on L). At ≤ 12 atoms there is no completion strength at which the quantum sampler wins on final route length, and we say so. **Where the register wins is per shot**: 60–100 % of its ground states are valid maximum packings against 3–6 % of random bitstrings on M and L, a 10–30× advantage in the quantity the device is asked for. **Raw feasibility tracks the embedding**: instances with 0 violable overlap edges give 100 % valid packings, one with 8 of 26 gives ~25 %. **The sweep beats the exact ground state** (M, first session: 6.5 % vs 12.8 % under the greedy completion) because it also samples low excited states and the pipeline keeps the best — the reason pulses are scored per shot, not by mean energy.

Tier S is degenerate for this model (candidates hold 1–2 customers). Under the exact completion the model gap equals the CVRP gap, since set partitioning with exact tour costs is the CVRP itself.

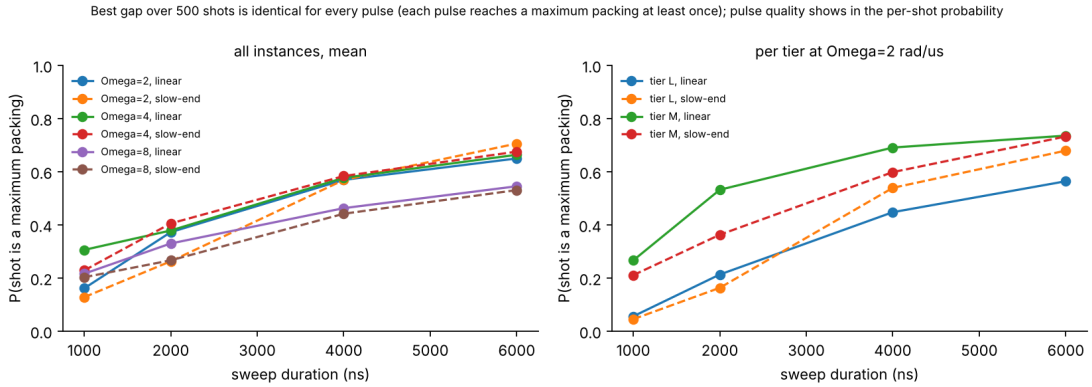


Figure 1: Pulse-shaping study: probability that a shot is a maximum packing versus sweep duration (left: all instances by $\Omega_{\max}$ and detuning shape; right: per tier at $\Omega_{\max} = 2$ rad/µs).

6. What did not work

- **Dense one-hot by blockade (8 / 30 / 60 atoms).** Tier S: the realised register's exact ground state decodes to the model optimum on four of four seeds (in-group coupling 55 rad/µs, $\delta_f \approx 25$). Tier M, $K = 3$ triangles of side 5 µm, block-centred fit of the 3×3 coupling blocks that an assignment actually sees: effective coupling error 0.97–0.98 for $\beta \in \{0, 1\}$; brute-forcing the realised energy over all 9842 clusterings, the realised minimiser is 57–253 % above the model optimum and the model optimum ranks 1349th–4770th. A K-gon per customer cannot be oriented for ~10 partners at once, and the $K = 4$ square has a 7.07 µm diagonal whose coupling is 1/8 of the side's ($\sqrt{2}^6 = 8$), so one-hot at $K = 4$ is marginal regardless.
- **Dense one-bit bisection (N atoms).** Exact at $N = 4$; at $N = 10$ and 15 every pair wants a coupling, the fit crams all atoms into an 8–10 µm disc (log-RMS coupling error 2–2.4) and the realised ground state is 25–55 % above the $K = 2$ optimum.
- **Hierarchical bisection on a sparse far-conflict graph** embeds cleanly at ≤ 2 edges per atom but is too weak a model: even the ideal MIS gives mean gaps of 10–16 % (M) and 21–24 % (L).
- **Richer candidate pools** (two fill levels per centre, grid centres) raise coverage on paper and destroy the embedding: overlap cliques, 23 of 51 edges violable, 0 % raw feasibility.
- **Petal (angular-window) candidates**, the shape of optimal routes around a central depot: coverage up (0.91), gap not (ideal 5.3 % vs 4.8 % for balls on the first 5 L instances), and mixed with balls the overlap graph gets 5–15 violable edges on 12 atoms.
- **Savings-weighted maximum-weight IS** (what the DMM would implement; `AnalogDevice` has no DMM, `DigitalAnalogDevice` has `dmm_0`): *worse* than uniform MIS even for a perfect device, L 8.7 % vs 4.8 %, M 5.6 % vs 2.7 %. Every candidate the device fixes is a constraint on the exact completion, and weighting fixes bigger ones. The DMM sequence was therefore not built.
- Not attempted: Track B.

7. Reproduction and AI use

`README.md` gives the one-command path (instances → references → classical and QuTiP runs → figures); `requirements.txt` and `report/pinned.txt` pin the environment. Claude was used for brainstorming, the derivations' first drafts, most of the code and this report's draft; every claim was re-run and every derivation re-done by a team member, as recorded in `AI_USE.md`.